

THE COMPLETE BIOLOGY AND MEDICINE ROADMAP

Zindagi ki Bunyaad se Nobel-Level Research aur Medical Practice tak

Oxford . Cambridge . Harvard Medical School . Johns Hopkins . Stanford Medicine

Yeh PDF Biology aur Medicine ka **mukammal ordered roadmap** hai. Bilkul shuruaat ('kya plants saans lete hain?') se lekar Oxford DPhil, Cambridge PhD, aur Harvard Medical School tak ke research frontiers tak. Isme **Cell Biology, Genetics, Evolution, Ecology, Human Anatomy, Physiology, Biochemistry, Microbiology, Immunology, Neuroscience, aur Clinical Medicine** -- sab kuch ordered sequence mein hai. Yeh roadmap un bacchon ke liye bhi hai jo doctor banna chahte hain, aur un ke liye bhi jo biological research mein Nobel Prize jeetna chahte hain.

LEVEL	STAGE	FOCUS
Level 1	Age 3-7	Living things, plants, animals, our bodies -- observation and wonder
Level 2	Age 7-11	Body systems, life cycles, ecosystems, health
Level 3	Age 11-14	Cells, genetics intro, evolution, ecology -- NCERT level
Level 4	Age 14-16	IGCSE Biology -- all core topics
Level 5	Age 16-18	A-Level / IB -- advanced biology, biochemistry
Level 6	Age 18-21	Undergraduate BSc -- molecular biology, physiology, medicine pre-clinical
Level 7	Age 21-24	Graduate / MBBS clinical / MSc research
Level 8	Age 24-27	Medical specialisation / PhD biology research
Level 9	Age 27+	Research frontier -- Nobel-level topics

CHAPTER 1 -- EARLY LIFE SCIENCE (Level 1 | Age 3-7)

1.1 Living and Non-Living Things

- 1.1.1 Living things: move, grow, feed, breathe, reproduce, feel, excrete
- 1.1.2 Non-living things: rock, water, car -- do not grow or reproduce
- 1.1.3 Sorting living and non-living things
- 1.1.4 Micro-organisms exist but are too small to see -- concept
- 1.1.5 All living things need food, water, and air

1.2 Plants

- 1.2.1 Parts of a plant: root, stem, leaf, flower, fruit, seed
- 1.2.2 What roots do: absorb water and anchor the plant
- 1.2.3 What leaves do: make food using sunlight
- 1.2.4 Plants need: sunlight, water, air, soil
- 1.2.5 Seeds grow into plants: germination concept
- 1.2.6 Different types of plants: trees, shrubs, herbs, creepers
- 1.2.7 Plants give us: food, oxygen, wood, medicine

1.3 Animals

- 1.3.1 Groups of animals: mammals, birds, reptiles, fish, amphibians, insects
- 1.3.2 Wild animals vs pet animals vs farm animals
- 1.3.3 What animals eat: herbivores, carnivores, omnivores
- 1.3.4 Animal homes: nest, burrow, den, hive, web
- 1.3.5 How animals move: walk, swim, fly, slither, jump
- 1.3.6 Baby animals and their parents
- 1.3.7 Animal sounds and how they communicate
- 1.3.8 Animals in different habitats: desert, forest, ocean, Arctic

1.4 Our Bodies

- 1.4.1 External body parts: head, neck, trunk, limbs
- 1.4.2 Senses: eyes (sight), ears (hearing), nose (smell), tongue (taste), skin (touch)
- 1.4.3 Teeth: types and care
- 1.4.4 Skeleton: bones hold us up
- 1.4.5 Heart: pumps blood
- 1.4.6 Lungs: we breathe air
- 1.4.7 Brain: controls everything
- 1.4.8 Healthy habits: food, sleep, exercise, hygiene

1.5 Environment and Seasons

- 1.5.1 Seasons and how they affect plants and animals
- 1.5.2 Food chain: sun -- plant -- animal -- another animal
- 1.5.3 Caring for the environment: not littering, saving water
- 1.5.4 Plants and animals depend on each other

** Parents: nature walks, growing a seed in a pot, observing ants -- hands-on is best at this age.*

CHAPTER 2 -- PRIMARY BIOLOGY (Level 2 | Age 7-11)

2.1 Cell Basics -- Introduction

- 2.1.1 All living things made of cells -- Robert Hooke
- 2.1.2 Cell is the basic unit of life
- 2.1.3 Unicellular vs multicellular organisms
- 2.1.4 Cell membrane, cytoplasm, nucleus -- basic
- 2.1.5 Plant cells have cell wall and chloroplasts
- 2.1.6 Using a microscope to see cells

2.2 Human Body Systems

- 2.2.1 Digestive system: mouth, oesophagus, stomach, small intestine, large intestine
- 2.2.2 What digestion does: breaks food into nutrients
- 2.2.3 Circulatory system: heart, blood, blood vessels
- 2.2.4 Blood carries oxygen and nutrients
- 2.2.5 Respiratory system: nose, trachea, lungs
- 2.2.6 Breathing: inhale oxygen, exhale carbon dioxide
- 2.2.7 Skeletal system: bones, joints, cartilage
- 2.2.8 Muscular system: muscles move bones
- 2.2.9 Nervous system: brain, spinal cord, nerves
- 2.2.10 Excretory system: kidneys, skin, lungs remove waste
- 2.2.11 Reproductive system: basic -- how babies are made (age-appropriate)

2.3 Plants -- Photosynthesis and Reproduction

- 2.3.1 Photosynthesis: sunlight + water + CO₂ -- glucose + oxygen
- 2.3.2 Chlorophyll: green pigment that captures light
- 2.3.3 Respiration in plants: opposite of photosynthesis at night
- 2.3.4 Pollination: transfer of pollen from anther to stigma
- 2.3.5 Fertilisation: pollen + ovule = seed
- 2.3.6 Seed dispersal: wind, water, animals, explosion
- 2.3.7 Germination conditions: water, warmth, oxygen
- 2.3.8 Vegetative reproduction: cuttings, runners, bulbs
- 2.3.9 Transpiration: water loss through leaves

2.4 Ecosystems and Food Webs

- 2.4.1 Habitat: where an organism lives
- 2.4.2 Population, community, ecosystem
- 2.4.3 Food chain: producer -- primary consumer -- secondary consumer
- 2.4.4 Food web: interconnected food chains
- 2.4.5 Producers: plants make their own food
- 2.4.6 Consumers: herbivores, carnivores, omnivores, decomposers
- 2.4.7 Decomposers: fungi and bacteria break down dead matter
- 2.4.8 Nutrient cycles: carbon cycle, nitrogen cycle -- basic
- 2.4.9 Adaptation: how organisms suit their environment
- 2.4.10 Predator and prey relationships
- 2.4.11 Pollution and its effects on ecosystems
- 2.4.12 Endangered species and conservation

2.5 Health and Disease -- Primary

- 2.5.1 Nutrients: carbohydrates, proteins, fats, vitamins, minerals, water
- 2.5.2 Balanced diet
- 2.5.3 Deficiency diseases: scurvy (Vit C), rickets (Vit D), anaemia (iron)
- 2.5.4 Infectious diseases: caused by pathogens

- 2.5.5 How infections spread: air, water, contact, vectors
- 2.5.6 Vaccines: training the immune system
- 2.5.7 Antibiotics: kill bacteria (not viruses)
- 2.5.8 Personal hygiene and public health
- 2.5.9 Drug awareness: legal and illegal, effects on body

2.6 Life Cycles and Reproduction

- 2.6.1 Life cycle of a flowering plant
- 2.6.2 Life cycle of a mammal
- 2.6.3 Life cycle of a butterfly: metamorphosis
- 2.6.4 Life cycle of a frog: amphibian metamorphosis
- 2.6.5 Sexual vs asexual reproduction
- 2.6.6 Puberty: physical and emotional changes
- 2.6.7 Human reproduction: fertilisation, pregnancy, birth -- age-appropriate

** NCERT Science Class 5-7 covers this level. Growing mung beans, dissecting a flower -- simple experiments.*

3.1 Cell Biology

- 3.1.1 Cell theory: all cells come from cells, all life from cells
- 3.1.2 Prokaryotic cells: no nucleus, bacteria
- 3.1.3 Eukaryotic cells: have nucleus, animals and plants
- 3.1.4 Animal cell organelles: nucleus, mitochondria, ribosomes, ER, Golgi, vacuoles, lysosomes
- 3.1.5 Plant cell organelles: all above + cell wall, chloroplasts, large central vacuole
- 3.1.6 Functions of each organelle
- 3.1.7 Cell specialisation: how cells become different
- 3.1.8 Stem cells: undifferentiated, potential
- 3.1.9 Tissues, organs, organ systems, organism -- levels
- 3.1.10 Transport across membranes: diffusion, osmosis, active transport
- 3.1.11 Osmosis: water potential, turgor, plasmolysis
- 3.1.12 Cell division: mitosis -- identical copies for growth
- 3.1.13 Cell division: meiosis -- for gametes, halves chromosome number
- 3.1.14 Cell cycle: interphase, prophase, metaphase, anaphase, telophase

3.2 Genetics -- Introduction

- 3.2.1 DNA: deoxyribonucleic acid, double helix, Watson and Crick
- 3.2.2 Chromosomes: packages of DNA in the nucleus
- 3.2.3 Humans have 46 chromosomes (23 pairs)
- 3.2.4 Genes: sections of DNA that code for proteins
- 3.2.5 Alleles: different versions of a gene
- 3.2.6 Dominant and recessive alleles
- 3.2.7 Homozygous and heterozygous
- 3.2.8 Genotype and phenotype
- 3.2.9 Monohybrid crosses: Punnett squares
- 3.2.10 Mendel's laws: segregation and independent assortment
- 3.2.11 Sex determination: XX female, XY male
- 3.2.12 Sex-linked inheritance: colour blindness, haemophilia
- 3.2.13 Genetic disorders: Down syndrome, cystic fibrosis, sickle cell
- 3.2.14 Mutation: change in DNA sequence
- 3.2.15 Causes of mutation: radiation, chemicals, errors in replication

3.3 Evolution

- 3.3.1 Charles Darwin: On the Origin of Species (1859)
- 3.3.2 Natural selection: variation, competition, survival, reproduction
- 3.3.3 Adaptation: suited to environment
- 3.3.4 Speciation: new species form over time
- 3.3.5 Evidence for evolution: fossils, comparative anatomy, DNA
- 3.3.6 Homologous structures: same origin, different function
- 3.3.7 Analogous structures: different origin, same function
- 3.3.8 Vestigial organs: reduced, no function
- 3.3.9 Antibiotic resistance as evolution in action
- 3.3.10 Classification: kingdom, phylum, class, order, family, genus, species
- 3.3.11 Binomial nomenclature: Homo sapiens
- 3.3.12 Five kingdoms: Monera, Protista, Fungi, Plantae, Animalia
- 3.3.13 Three domains: Bacteria, Archaea, Eukarya

3.4 Ecology

- 3.4.1 Biotic and abiotic factors
- 3.4.2 Limiting factors: food, space, disease, predation

- 3.4.3 Population growth: exponential and logistic (S-curve)
- 3.4.4 Carrying capacity
- 3.4.5 Interspecific competition, intraspecific competition
- 3.4.6 Predator-prey cycles: Lotka-Volterra -- concept
- 3.4.7 Symbiosis: mutualism, commensalism, parasitism
- 3.4.8 Succession: primary and secondary
- 3.4.9 Carbon cycle: detailed
- 3.4.10 Nitrogen cycle: fixation, nitrification, denitrification
- 3.4.11 Water cycle: evaporation, transpiration, precipitation
- 3.4.12 Biomes of the world: detailed
- 3.4.13 Human impact: deforestation, habitat loss, pollution, climate change
- 3.4.14 Biodiversity: importance and measurement
- 3.4.15 Conservation strategies: in situ, ex situ

3.5 Human Biology -- Body Systems

- 3.5.1 Digestive system: detailed -- enzymes, absorption, villi
- 3.5.2 Digestive enzymes: amylase, protease, lipase -- where made, what they do
- 3.5.3 Circulatory system: double circulation, heart structure
- 3.5.4 Heartbeat: SA node, AV node, Purkinje fibres
- 3.5.5 Blood: red cells, white cells, platelets, plasma -- functions
- 3.5.6 Blood vessels: arteries, veins, capillaries -- structure and function
- 3.5.7 Lymphatic system: lymph, lymph nodes, spleen
- 3.5.8 Respiratory system: ventilation, gas exchange in alveoli
- 3.5.9 Breathing mechanism: diaphragm, intercostal muscles
- 3.5.10 Nervous system: central vs peripheral, somatic vs autonomic
- 3.5.11 Neuron: structure, impulse transmission, synapse, neurotransmitters
- 3.5.12 Reflex arc
- 3.5.13 Brain regions: cerebrum, cerebellum, medulla oblongata, hypothalamus
- 3.5.14 Endocrine system: hormones, glands
- 3.5.15 Key hormones: insulin, glucagon, adrenaline, oestrogen, testosterone, ADH, FSH, LH
- 3.5.16 Homeostasis: blood glucose, temperature, water balance
- 3.5.17 Kidney: structure, ultrafiltration, selective reabsorption, osmoregulation
- 3.5.18 Skin: thermoregulation -- sweating, vasodilation, vasoconstriction
- 3.5.19 Immune system: non-specific and specific defence
- 3.5.20 Specific immune response: B cells, T cells, antibodies, memory cells
- 3.5.21 Active and passive immunity
- 3.5.22 Musculoskeletal system: bone structure, joints, antagonistic muscles
- 3.5.23 Reproductive system: male and female anatomy
- 3.5.24 Menstrual cycle: hormonal control
- 3.5.25 Pregnancy and fetal development
- 3.5.26 Contraception: methods and mechanisms

** NCERT Biology Class 9-10 and Cambridge IGCSE Biology (0610) past papers -- essential practice.*

4.1 Biochemistry -- Introduction

- 4.1.1 Biological molecules: carbohydrates, lipids, proteins, nucleic acids, water
- 4.1.2 Carbohydrates: monosaccharides (glucose), disaccharides (sucrose), polysaccharides (starch, glycogen, cellulose)
- 4.1.3 Lipids: triglycerides, phospholipids, steroids
- 4.1.4 Proteins: amino acids, peptide bonds, primary to quaternary structure
- 4.1.5 Enzymes: biological catalysts, active site, substrate
- 4.1.6 Enzyme kinetics: Michaelis-Menten -- Km and Vmax concept
- 4.1.7 Factors affecting enzyme activity: temperature, pH, substrate concentration, inhibitors
- 4.1.8 Enzyme inhibition: competitive and non-competitive
- 4.1.9 Nucleic acids: DNA and RNA structure
- 4.1.10 DNA double helix: base pairing (A-T, G-C)
- 4.1.11 DNA replication: semi-conservative, helicase, DNA polymerase
- 4.1.12 Protein synthesis: transcription (DNA to mRNA) and translation (mRNA to protein)
- 4.1.13 Codons and the genetic code
- 4.1.14 tRNA and ribosomes in translation
- 4.1.15 Water: hydrogen bonds, properties, biological importance

4.2 Cell Biology -- Advanced

- 4.2.1 Fluid mosaic model of cell membrane
- 4.2.2 Phospholipid bilayer: structure and function
- 4.2.3 Membrane proteins: integral, peripheral, glycoproteins, channel proteins
- 4.2.4 Endocytosis and exocytosis
- 4.2.5 Cell signalling: receptors, second messengers -- concept
- 4.2.6 Mitosis: detailed stages -- PMAT
- 4.2.7 Meiosis: detailed stages, crossing over, chiasmata
- 4.2.8 Significance of meiosis: genetic variation
- 4.2.9 Cancer: uncontrolled cell division, tumours, oncogenes, tumour suppressors

4.3 Molecular Genetics

- 4.3.1 Gene expression: regulation -- operons in prokaryotes (lac operon)
- 4.3.2 Epigenetics: methylation, histone modification -- concept
- 4.3.3 Genetic engineering: restriction enzymes, ligase, vectors, transformation
- 4.3.4 Recombinant DNA technology
- 4.3.5 Polymerase Chain Reaction (PCR)
- 4.3.6 Gel electrophoresis: separating DNA fragments by size
- 4.3.7 Gene cloning: in bacteria
- 4.3.8 Genetically modified organisms (GMOs): Bt crops, golden rice
- 4.3.9 CRISPR-Cas9: gene editing -- concept
- 4.3.10 DNA profiling (fingerprinting): forensics, paternity
- 4.3.11 Genome sequencing: Human Genome Project
- 4.3.12 Stem cell therapy: embryonic and adult stem cells

4.4 Physiology -- Advanced

- 4.4.1 Photosynthesis: light-dependent and light-independent reactions
- 4.4.2 Light-dependent: chlorophyll, photosystems I and II, ATP synthesis, NADPH
- 4.4.3 Light-independent (Calvin cycle): CO₂ fixation, G3P, RuBisCO
- 4.4.4 Factors limiting photosynthesis: light, CO₂, temperature
- 4.4.5 Cellular respiration: aerobic and anaerobic
- 4.4.6 Glycolysis: glucose to pyruvate, ATP and NADH
- 4.4.7 Krebs cycle (citric acid cycle): pyruvate to CO₂, NADH, FADH₂

- 4.4.8 Electron transport chain: chemiosmosis, 36-38 ATP
- 4.4.9 Anaerobic respiration: lactic acid (animals), ethanol + CO₂ (yeast)
- 4.4.10 Respiratory quotient (RQ)
- 4.4.11 Transport in plants: xylem (water/minerals), phloem (sugars)
- 4.4.12 Transpiration: stomata, guard cells, factors affecting rate
- 4.4.13 Translocation: source to sink

4.5 Microbiology

- 4.5.1 Viruses: structure, replication cycle, lytic and lysogenic
- 4.5.2 Bacteria: structure, Gram staining, reproduction
- 4.5.3 Fungi: structure, reproduction, importance
- 4.5.4 Protists: amoeba, paramecium, plasmodium (malaria)
- 4.5.5 Pathogens and diseases: bacterial (tuberculosis, cholera), viral (HIV, influenza), fungal, protist
- 4.5.6 Transmission: direct contact, droplets, vectors, food and water, fomites
- 4.5.7 Koch's postulates
- 4.5.8 Antibiotic resistance: mechanisms and implications
- 4.5.9 Vaccines: types (live attenuated, killed, subunit, mRNA)
- 4.5.10 Herd immunity: threshold
- 4.5.11 Aseptic technique
- 4.5.12 Culturing bacteria: agar plates, streak plate method

4.6 Ecology -- Advanced IGCSE

- 4.6.1 Energy flow in ecosystems: GPP, NPP, efficiency
- 4.6.2 Trophic levels and energy pyramids
- 4.6.3 Pyramids of numbers, biomass, energy
- 4.6.4 Ecological sampling: quadrats, transects, mark-recapture
- 4.6.5 Species diversity indices: Simpson's index
- 4.6.6 Climate change: greenhouse effect, feedback loops
- 4.6.7 Acid rain: formation, effects
- 4.6.8 Eutrophication: nutrient pollution in water
- 4.6.9 Bioaccumulation and biomagnification
- 4.6.10 Sustainable development and conservation

4.7 Animal Diversity and Anatomy

- 4.7.1 Animal phyla: Porifera, Cnidaria, Platyhelminthes, Annelida, Mollusca, Arthropoda, Echinodermata, Chordata
- 4.7.2 Vertebrate classes: fish, amphibians, reptiles, birds, mammals
- 4.7.3 Comparative anatomy: circulatory, respiratory, nervous systems across vertebrates
- 4.7.4 Insect structure and adaptations
- 4.7.5 Animal behaviour: innate and learned
- 4.7.6 Courtship, migration, communication in animals

** Cambridge IGCSE Biology (0610), Edexcel IGCSE -- extensive past paper practice essential. Draw and label diagrams.*

5.1 Advanced Biochemistry

- 5.1.1 Carbohydrate biochemistry: glycosidic bonds, reducing sugars, Benedict's test
- 5.1.2 Lipid biochemistry: esterification, saponification, HDL vs LDL
- 5.1.3 Protein structure: alpha helix, beta pleated sheet, disulfide bonds
- 5.1.4 Protein folding and denaturation
- 5.1.5 Enzyme kinetics: Lineweaver-Burk plot, allosteric regulation
- 5.1.6 Coenzymes and cofactors: NAD, FAD, coenzyme A
- 5.1.7 ATP: structure, synthesis, hydrolysis, role as energy currency
- 5.1.8 Vitamins and their biochemical roles
- 5.1.9 Biuret test, iodine test, emulsion test -- practicals

5.2 Advanced Cell Biology

- 5.2.1 Endomembrane system: ER, Golgi, vesicle trafficking
- 5.2.2 Mitochondria: structure and chemiosmosis in detail
- 5.2.3 Chloroplast: structure and photosystems in detail
- 5.2.4 Cytoskeleton: microfilaments, microtubules, intermediate filaments
- 5.2.5 Cell signalling pathways: receptor tyrosine kinases, cAMP
- 5.2.6 Apoptosis: programmed cell death, caspases
- 5.2.7 Cell cycle control: cyclins, CDKs, checkpoints
- 5.2.8 Oncogenes: ras, myc -- gain of function mutations
- 5.2.9 Tumour suppressor genes: p53, Rb
- 5.2.10 Stem cells: embryonic, adult, iPSCs

5.3 Advanced Genetics and Molecular Biology

- 5.3.1 Dihybrid crosses: independent assortment, 9:3:3:1 ratio
- 5.3.2 Linkage and crossing over: recombination frequencies
- 5.3.3 Chi-squared test for genetic data
- 5.3.4 Chromosome mapping
- 5.3.5 Epistasis: gene interactions affecting phenotype ratio
- 5.3.6 Quantitative genetics: polygenic inheritance, continuous variation
- 5.3.7 Epigenetics: detailed -- chromatin remodelling, imprinting
- 5.3.8 Non-coding RNA: siRNA, miRNA, lncRNA
- 5.3.9 Eukaryotic gene regulation: enhancers, promoters, transcription factors
- 5.3.10 DNA repair mechanisms: base excision, nucleotide excision, mismatch repair
- 5.3.11 Telomeres and telomerase: ageing
- 5.3.12 Transposons: jumping genes -- Barbara McClintock
- 5.3.13 Comparative genomics
- 5.3.14 Proteomics: protein expression and interaction
- 5.3.15 Metagenomics

5.4 Advanced Physiology

- 5.4.1 Photosynthesis: Z-scheme, cyclic and non-cyclic photophosphorylation
- 5.4.2 Calvin cycle: detailed -- RuBisCO, 3-PGA, G3P, RuBP regeneration
- 5.4.3 C₄ and CAM photosynthesis: adaptations to hot/dry environments
- 5.4.4 Respiration: detailed electron transport, complexes I-IV, ATP synthase
- 5.4.5 Substrate-level vs oxidative phosphorylation
- 5.4.6 Metabolic pathways: anabolism and catabolism
- 5.4.7 Haemoglobin: oxygen dissociation curve, Bohr effect, cooperative binding
- 5.4.8 Fetal haemoglobin vs adult haemoglobin
- 5.4.9 Blood clotting: coagulation cascade
- 5.4.10 Action potential: resting potential, depolarisation, repolarisation

- 5.4.11 Saltatory conduction in myelinated neurons
- 5.4.12 Synaptic transmission: neurotransmitters, receptors, summation
- 5.4.13 Hormonal control: negative and positive feedback
- 5.4.14 Insulin and glucagon: detailed mechanism, diabetes type 1 vs 2
- 5.4.15 ADH and osmoregulation
- 5.4.16 Muscle contraction: sliding filament theory, actin, myosin, tropomyosin
- 5.4.17 Neuromuscular junction

5.5 Evolution and Biodiversity -- Advanced

- 5.5.1 Population genetics: Hardy-Weinberg equilibrium
- 5.5.2 Mechanisms of evolution: natural selection, genetic drift, gene flow, mutation
- 5.5.3 Directional, stabilising, disruptive selection
- 5.5.4 Sexual selection: Fisherian runaway, honest signal
- 5.5.5 Speciation: allopatric and sympatric
- 5.5.6 Reproductive isolation: pre-zygotic and post-zygotic
- 5.5.7 Phylogenetics: cladistics, molecular phylogeny
- 5.5.8 Molecular clocks
- 5.5.9 Mass extinctions: five major events
- 5.5.10 Coevolution: host-pathogen, plant-pollinator
- 5.5.11 Kin selection and inclusive fitness: Hamilton's rule
- 5.5.12 Reciprocal altruism: Trivers
- 5.5.13 Sociobiology and evolutionary psychology

5.6 Neuroscience -- A-Level

- 5.6.1 Brain structure: lobes of cerebral cortex and their functions
- 5.6.2 Limbic system: amygdala, hippocampus, hypothalamus
- 5.6.3 Cerebellum and motor control
- 5.6.4 Brain imaging: fMRI, PET, EEG
- 5.6.5 Learning and memory: long-term potentiation (LTP)
- 5.6.6 Neurotransmitters: dopamine, serotonin, acetylcholine, noradrenaline, GABA, glutamate
- 5.6.7 Drugs and neurotransmitter systems
- 5.6.8 Neurological and psychiatric conditions: Alzheimer's, Parkinson's, depression, schizophrenia -- biological basis

** IB Biology HL, Cambridge A-Level Biology 9700 -- core texts: Campbell Biology, Alberts Molecular Biology of the Cell.*

6.1 Advanced Biochemistry

- 6.1.1 Enzyme mechanisms: serine proteases, acid-base catalysis
- 6.1.2 Metabolic regulation: allosteric enzymes, feedback inhibition
- 6.1.3 Glycolysis regulation: PFK-1, hexokinase
- 6.1.4 Gluconeogenesis: synthesis of glucose
- 6.1.5 Glycogen metabolism: glycogen synthase, phosphorylase
- 6.1.6 Pentose phosphate pathway: NADPH, nucleotide synthesis
- 6.1.7 Fatty acid oxidation (beta-oxidation)
- 6.1.8 Fatty acid synthesis
- 6.1.9 Ketogenesis and ketolysis
- 6.1.10 Amino acid catabolism: transamination, urea cycle
- 6.1.11 Purine and pyrimidine synthesis and salvage
- 6.1.12 Cholesterol synthesis: HMG-CoA reductase (statin target)
- 6.1.13 Lipoproteins: VLDL, LDL, HDL -- metabolism and disease
- 6.1.14 Vitamins: fat-soluble (A, D, E, K) and water-soluble (B vitamins, C)
- 6.1.15 Signal transduction: detailed pathways -- MAPK, PI3K/Akt, JAK-STAT

6.2 Molecular Biology and Genetics

- 6.2.1 DNA replication machinery: origin of replication, primase, DNA Pol I/III, ligase, topoisomerases
- 6.2.2 Chromatin structure: nucleosomes, histones, chromatin compaction
- 6.2.3 Transcription machinery: RNA Pol II, GTFs, mediator complex
- 6.2.4 RNA processing: 5' capping, polyadenylation, splicing (spliceosome)
- 6.2.5 Alternative splicing and its role in proteome diversity
- 6.2.6 Translation: initiation, elongation, termination
- 6.2.7 Post-translational modifications: phosphorylation, glycosylation, ubiquitination
- 6.2.8 Protein degradation: proteasome, ubiquitin-proteasome pathway
- 6.2.9 Unfolded protein response
- 6.2.10 Genome organisation: centromeres, telomeres, repetitive elements
- 6.2.11 Human genetics: GWAS, linkage analysis, SNPs
- 6.2.12 Inherited diseases: molecular mechanisms -- CFTR, dystrophin, BRCA1/2
- 6.2.13 Genomic medicine: pharmacogenomics, personalised medicine
- 6.2.14 Next-generation sequencing (NGS) technologies
- 6.2.15 Single-cell sequencing

6.3 Cell Biology -- Advanced

- 6.3.1 Membrane trafficking: SNARE proteins, Rab GTPases, clathrin
- 6.3.2 Autophagy: mechanism and role in disease
- 6.3.3 Mitosis regulation: MPF, APC/C
- 6.3.4 DNA damage response: ATM, ATR, CHK1/2, p53 pathway
- 6.3.5 Senescence: cellular ageing, SASP
- 6.3.6 Cancer biology: hallmarks of cancer (Hanahan and Weinberg)
- 6.3.7 Tumour microenvironment
- 6.3.8 Metastasis: EMT, invasion, colonisation
- 6.3.9 Angiogenesis: VEGF, tumour vasculature
- 6.3.10 Cancer immunology: immune evasion
- 6.3.11 Extracellular matrix: collagen, fibronectin, integrins

6.4 Anatomy and Histology

- 6.4.1 Anatomical terminology: planes, directions, regions
- 6.4.2 Integumentary system: skin layers, appendages

- 6.4.3 Skeletal system: bone histology, ossification, joints
- 6.4.4 Muscular system: skeletal, cardiac, smooth -- histology
- 6.4.5 Cardiovascular anatomy: heart chambers, valves, coronary circulation
- 6.4.6 Arterial and venous systems: major vessels
- 6.4.7 Lymphatic system anatomy
- 6.4.8 Respiratory anatomy: airways, lung structure, pleura
- 6.4.9 Digestive system anatomy: detailed
- 6.4.10 Liver anatomy and histology: lobules, hepatocytes
- 6.4.11 Pancreas: endocrine and exocrine
- 6.4.12 Urinary system: kidney anatomy, nephron structure
- 6.4.13 Male and female reproductive anatomy
- 6.4.14 Nervous system anatomy: cranial nerves, spinal cord
- 6.4.15 Endocrine glands: pituitary, thyroid, adrenal, gonads
- 6.4.16 Head and neck anatomy: clinical relevance

6.5 Physiology -- Systems Level

- 6.5.1 Cardiac physiology: Frank-Starling law, cardiac output, Starling forces
- 6.5.2 ECG: P wave, QRS complex, T wave, intervals
- 6.5.3 Blood pressure regulation: baroreceptors, RAAS, natriuretic peptides
- 6.5.4 Respiratory physiology: compliance, resistance, spirometry values
- 6.5.5 Ventilation/perfusion matching (V/Q)
- 6.5.6 Gas exchange: Fick's law, oxygen and CO₂ transport
- 6.5.7 Renal physiology: GFR, tubular secretion, concentration of urine
- 6.5.8 Acid-base balance: Henderson-Hasselbalch, buffers, compensation
- 6.5.9 Gastrointestinal physiology: motility, secretion, absorption
- 6.5.10 Hepatic physiology: metabolism, bile, detoxification
- 6.5.11 Endocrine physiology: HPG, HPA, HPT axes
- 6.5.12 Reproductive physiology: menstrual cycle hormones, spermatogenesis
- 6.5.13 Exercise physiology: cardiovascular and respiratory adaptations
- 6.5.14 Thermoregulation: hypothalamus, fever

6.6 Immunology

- 6.6.1 Innate immunity: physical barriers, pattern recognition receptors (PRRs), TLRs
- 6.6.2 Complement system: classical, alternative, lectin pathways
- 6.6.3 Phagocytosis: neutrophils, macrophages
- 6.6.4 Inflammation: mediators, cardinal signs, acute and chronic
- 6.6.5 Adaptive immunity: lymphocytes, antigen presentation
- 6.6.6 MHC class I and II: which cells express, what they present
- 6.6.7 T cell activation: TCR, CD4, CD8, costimulation
- 6.6.8 T helper subsets: Th1, Th2, Th17, Treg
- 6.6.9 Cytotoxic T cells: killing mechanisms
- 6.6.10 B cell activation and antibody production
- 6.6.11 Immunoglobulin classes: IgG, IgM, IgA, IgE, IgD -- roles
- 6.6.12 Antibody structure and function: Fab, Fc regions
- 6.6.13 Memory cells and vaccination
- 6.6.14 Hypersensitivity reactions: Types I-IV (Coombs and Gell)
- 6.6.15 Autoimmunity: mechanisms and diseases
- 6.6.16 Immunodeficiency: primary and secondary (HIV/AIDS in detail)
- 6.6.17 Transplant immunology: HLA matching, rejection, immunosuppression
- 6.6.18 Tumour immunology: checkpoint inhibitors, CAR-T cells

6.7 Microbiology and Infectious Disease

- 6.7.1 Bacterial pathogenesis: virulence factors, toxins, adhesins

- 6.7.2 Mechanisms of antibiotic action: cell wall, protein synthesis, DNA gyrase, membrane
- 6.7.3 Antibiotic resistance mechanisms: beta-lactamase, efflux pumps, target modification
- 6.7.4 Viral replication cycles: detailed
- 6.7.5 HIV: life cycle, ART targets, AIDS progression
- 6.7.6 Hepatitis B and C: epidemiology, treatment
- 6.7.7 Respiratory infections: influenza, COVID-19 -- biology
- 6.7.8 Malaria: Plasmodium life cycle, treatment
- 6.7.9 Tuberculosis: Mycobacterium, latency, MDR-TB
- 6.7.10 Microbiome: gut flora, dysbiosis, role in health
- 6.7.11 Prions: misfolded proteins, CJD
- 6.7.12 Epidemiology: R0, herd immunity, outbreak investigation
- 6.7.13 Hospital-acquired infections (HAIs) and infection control

6.8 Pharmacology -- Foundations

- 6.8.1 Pharmacokinetics: absorption, distribution, metabolism, excretion (ADME)
- 6.8.2 Drug bioavailability and first-pass effect
- 6.8.3 Volume of distribution, clearance, half-life
- 6.8.4 Pharmacodynamics: dose-response curves, EC50, therapeutic index
- 6.8.5 Drug receptors: agonists, antagonists, partial agonists, inverse agonists
- 6.8.6 G-protein coupled receptors (GPCRs): signalling
- 6.8.7 Ion channel pharmacology
- 6.8.8 Enzyme inhibitors as drugs: statins, ACE inhibitors
- 6.8.9 Drug interactions: CYP450 enzyme system
- 6.8.10 Adverse drug reactions: types A, B, C, D
- 6.8.11 Classes of drugs: cardiovascular, CNS, antimicrobial, anti-inflammatory
- 6.8.12 Autonomic pharmacology: sympathomimetics, beta-blockers, anticholinergics
- 6.8.13 Drug development: preclinical, Phase I-IV trials

* *Guyton & Hall (Physiology), Robbins (Pathology), Goodman & Gilman (Pharmacology), Gray's Anatomy -- the classic texts.*

7.1 Clinical Medicine -- Core

- 7.1.1 History taking: presenting complaint, HPC, PMH, DH, FH, SH, ROS
- 7.1.2 Clinical examination: general, cardiovascular, respiratory, abdominal, neurological
- 7.1.3 Diagnosis: differential diagnosis, clinical reasoning
- 7.1.4 Investigations: blood tests (FBC, U&E, LFTs, TFTs, ABG), imaging (CXR, CT, MRI, USS), histology
- 7.1.5 Evidence-based medicine: RCTs, systematic reviews, meta-analysis
- 7.1.6 Clinical guidelines: NICE, WHO, national protocols

7.2 Pathology

- 7.2.1 Cellular adaptations: hypertrophy, hyperplasia, atrophy, metaplasia
- 7.2.2 Cell injury and necrosis: ischaemic, coagulative, liquefactive, caseous
- 7.2.3 Inflammation: acute and chronic in detail
- 7.2.4 Wound healing: primary and secondary intention
- 7.2.5 Haemodynamic disorders: oedema, hyperaemia, thrombosis, embolism, infarction, shock
- 7.2.6 Neoplasia: classification, grading, staging (TNM), carcinogenesis
- 7.2.7 Haematological disorders: anaemias, leukaemias, lymphomas
- 7.2.8 Cardiovascular pathology: atherosclerosis, MI, heart failure
- 7.2.9 Respiratory pathology: pneumonia, COPD, asthma, lung cancer
- 7.2.10 Gastrointestinal pathology: peptic ulcer, IBD, liver cirrhosis, colorectal cancer
- 7.2.11 Renal pathology: glomerulonephritis, nephrotic syndrome, AKI, CKD
- 7.2.12 Endocrine pathology: diabetes, thyroid disorders, Cushing's, Addison's
- 7.2.13 Neurological pathology: stroke, dementia, brain tumours, MS
- 7.2.14 Musculoskeletal pathology: osteoporosis, arthritis, bone tumours
- 7.2.15 Dermatological pathology: common skin conditions, melanoma

7.3 Clinical Specialties Overview

- 7.3.1 Cardiology: ECG interpretation, heart failure, arrhythmias, interventions
- 7.3.2 Respiratory medicine: spirometry, asthma management, COPD, PE
- 7.3.3 Gastroenterology: endoscopy, IBD, hepatology, GI bleed
- 7.3.4 Nephrology: renal replacement therapy, AKI vs CKD management
- 7.3.5 Neurology: stroke, epilepsy, headache, movement disorders
- 7.3.6 Endocrinology: diabetes management, thyroid, adrenal
- 7.3.7 Haematology: blood disorders, bone marrow, coagulation
- 7.3.8 Oncology: cancer treatment -- surgery, chemotherapy, radiotherapy, immunotherapy
- 7.3.9 Infectious diseases: tropical medicine, HIV management, sepsis
- 7.3.10 Rheumatology: autoimmune diseases, biologics
- 7.3.11 Surgery: general principles, common operations
- 7.3.12 Obstetrics and gynaecology: antenatal care, labour, gynaecological conditions
- 7.3.13 Paediatrics: child development, common paediatric conditions
- 7.3.14 Psychiatry: mental health conditions, psychopharmacology, psychotherapy
- 7.3.15 Emergency medicine: ABCDE approach, trauma, critical care

7.4 Research Methods in Biology and Medicine

- 7.4.1 Study designs: RCT, cohort, case-control, cross-sectional, case series
- 7.4.2 Bias and confounding: types, control
- 7.4.3 Statistical methods: t-tests, chi-squared, ANOVA, regression
- 7.4.4 Survival analysis: Kaplan-Meier, hazard ratio
- 7.4.5 Diagnostic test statistics: sensitivity, specificity, PPV, NPV, ROC
- 7.4.6 Systematic review and meta-analysis: PRISMA
- 7.4.7 Research ethics: Declaration of Helsinki, Belmont Report

- 7.4.8 Informed consent and research governance
- 7.4.9 Laboratory techniques: Western blot, ELISA, flow cytometry, immunofluorescence
- 7.4.10 Animal models in research: 3Rs principle
- 7.4.11 Clinical trial design: Phase I-IV, adaptive trials
- 7.4.12 Translational medicine: bench to bedside
- 7.4.13 Bioinformatics: sequence analysis, pathway analysis

7.5 Neuroscience -- Graduate Level

- 7.5.1 Synaptic plasticity: LTP, LTD, STDP
- 7.5.2 Neurotransmitter systems: detailed pharmacology
- 7.5.3 Neural circuits: basal ganglia, cerebellum, cortex
- 7.5.4 Sensory neuroscience: visual cortex, somatosensory maps
- 7.5.5 Motor neuroscience: motor cortex, descending pathways
- 7.5.6 Sleep and circadian rhythms: biological clock
- 7.5.7 Pain: nociception, central sensitisation, chronic pain
- 7.5.8 Addiction: reward pathways, dopamine
- 7.5.9 Neurodegeneration: Alzheimer's, Parkinson's, ALS -- mechanisms
- 7.5.10 Neurogenesis: adult neurogenesis in hippocampus
- 7.5.11 Brain-computer interfaces: fundamentals
- 7.5.12 Computational neuroscience: Hodgkin-Huxley model

** Kumar & Clark (Clinical Medicine), Harrison's (Internal Medicine), Oxford Handbook of Clinical Medicine -- essential clinical references.*

8.1 Advanced Molecular Biology and Genomics

- 8.1.1 CRISPR-Cas9: mechanism, PAM sequence, guide RNA, HDR vs NHEJ repair
- 8.1.2 Base editing and prime editing
- 8.1.3 Epigenome editing: CRISPRa, CRISPRi
- 8.1.4 Long-read sequencing: Oxford Nanopore, PacBio
- 8.1.5 Spatial transcriptomics
- 8.1.6 Multi-omics integration: genomics, transcriptomics, proteomics, metabolomics
- 8.1.7 Structural biology: X-ray crystallography, cryo-EM, NMR for proteins
- 8.1.8 Protein structure prediction: AlphaFold2 and its implications
- 8.1.9 Synthetic biology: genetic circuits, BioBricks, synthetic genomes
- 8.1.10 Gene therapy: viral and non-viral vectors, clinical applications

8.2 Advanced Immunology and Oncology

- 8.2.1 Immunotherapy: checkpoint inhibitors (PD-1/PD-L1, CTLA-4)
- 8.2.2 CAR-T cell therapy: design, mechanism, clinical use
- 8.2.3 Bispecific antibodies
- 8.2.4 Tumour microenvironment: immunosuppression, myeloid cells
- 8.2.5 Cancer immunoediting: elimination, equilibrium, escape
- 8.2.6 Liquid biopsy: circulating tumour DNA (ctDNA)
- 8.2.7 Cancer genomics: mutation signatures, clonal evolution
- 8.2.8 Precision oncology: targeted therapies, EGFR, ALK inhibitors
- 8.2.9 Onco-immunology: innate immune cells in cancer
- 8.2.10 mRNA vaccines: mechanism, design, COVID-19 lessons

8.3 Developmental Biology and Regenerative Medicine

- 8.3.1 Embryonic development: cleavage, gastrulation, neurulation
- 8.3.2 Pattern formation: morphogen gradients, Hox genes
- 8.3.3 Cell fate determination: transcription factor networks
- 8.3.4 Induced pluripotent stem cells (iPSCs): Yamanaka factors
- 8.3.5 Organoids: growing organs in a dish
- 8.3.6 Tissue engineering: scaffolds, bioprinting
- 8.3.7 Regeneration: model organisms (zebrafish, salamander)
- 8.3.8 Ageing biology: hallmarks of ageing, senolytics
- 8.3.9 Epigenetics of ageing: clocks (Horvath)

8.4 Systems Biology and Computational Biology

- 8.4.1 Network biology: protein-protein interaction networks
- 8.4.2 Metabolic flux analysis
- 8.4.3 Gene regulatory networks: Boolean models
- 8.4.4 Machine learning in biology: protein function prediction
- 8.4.5 Deep learning for genomics: splicing prediction, regulatory elements
- 8.4.6 Drug target identification using AI
- 8.4.7 Pharmacokinetic/pharmacodynamic (PK/PD) modelling
- 8.4.8 Population genetics modelling
- 8.4.9 Ecological modelling: species distribution, climate change impact

8.5 Global Health and Public Health

- 8.5.1 Epidemiology advanced: surveillance, outbreak response
- 8.5.2 Pandemic preparedness: lessons from COVID-19, Ebola, flu
- 8.5.3 One Health: human-animal-environment interface
- 8.5.4 Non-communicable diseases: global burden, prevention

8.5.5 Health systems: Beveridge, Bismarck, private models

8.5.6 Social determinants of health

8.5.7 Global mental health

8.5.8 Neglected tropical diseases

8.5.9 Antimicrobial resistance: global threat, One Health approach

8.5.10 Health economics: QALYs, cost-effectiveness, NICE methodology

CHAPTER 9 -- RESEARCH FRONTIER (Level 9 | Nobel | Level Topics)

Yeh wo areas hain jahan Nobel Prizes, Lasker Awards, aur Breakthrough Prizes milte hain. Oxford Dunn School, Cambridge MRC-LMB, Harvard Medical School, Salk Institute, Broad Institute, Wellcome Sanger Institute -- yahan duniya ki top biological research hoti hai.

9.1 RNA Biology Revolution

- 9.1.1 RNA splicing: spliceosome structure -- 2009 Nobel (Ramakrishnan, Steitz, Yonath inspired)
- 9.1.2 Small non-coding RNAs: miRNA, siRNA, piRNA mechanisms
- 9.1.3 lncRNA: regulation of gene expression
- 9.1.4 Circular RNAs: back-splicing, functions
- 9.1.5 RNA modifications: m6A, pseudouridine -- epitranscriptomics
- 9.1.6 RNA-protein interactions: CLIP-seq
- 9.1.7 mRNA therapeutics: beyond COVID vaccines
- 9.1.8 RNA targeting with CRISPR: RCas9

9.2 CRISPR and Gene Editing Frontier

- 9.2.1 2020 Nobel Prize in Chemistry: Doudna and Charpentier
- 9.2.2 CRISPR screens: genome-wide functional screens
- 9.2.3 Prime editing: search and replace for genomes
- 9.2.4 In vivo CRISPR delivery: lipid nanoparticles, AAV
- 9.2.5 First CRISPR therapies: sickle cell disease, beta-thalassaemia (2023)
- 9.2.6 Epigenome editing at scale
- 9.2.7 Ethical debates: germline editing, He Jiankui case

9.3 Structural Biology and AlphaFold

- 9.3.1 Cryo-EM revolution: 2017 Nobel Prize (Dubochet, Frank, Henderson)
- 9.3.2 Atomic force microscopy (AFM) for biology
- 9.3.3 AlphaFold2: solving the protein folding problem
- 9.3.4 ESMFold and other AI structure predictors
- 9.3.5 Protein design: RFDiffusion, ProteinMPNN
- 9.3.6 De novo protein design for therapeutics
- 9.3.7 Structural vaccinology: structure-based vaccine design

9.4 Cell Biology Frontiers

- 9.4.1 Phase separation: liquid-liquid phase separation in biology
- 9.4.2 Membraneless organelles: P-bodies, stress granules
- 9.4.3 Biomolecular condensates: Rosen, Bhatt labs
- 9.4.4 Autophagy: 2016 Nobel Prize (Ohsumi)
- 9.4.5 Organelle communication: ER-mitochondria contact sites
- 9.4.6 Mitochondrial dynamics: fusion, fission
- 9.4.7 Cellular senescence: causes, consequences, senolytics
- 9.4.8 Mechanical biology: how cells sense and respond to forces

9.5 Neuroscience Frontier

- 9.5.1 Connectomics: mapping every synapse -- fly brain completed 2023
- 9.5.2 Spatial transcriptomics of the brain
- 9.5.3 Optogenetics: 2013 Breakthrough Prize (Boyden, Deisseroth)
- 9.5.4 Expansion microscopy
- 9.5.5 Brain organoids: modelling neurological disease
- 9.5.6 Neural basis of consciousness: free energy principle (Friston)
- 9.5.7 Psychedelics as therapeutics: psilocybin, ketamine
- 9.5.8 Neuralink and brain-computer interfaces
- 9.5.9 Alzheimer's: amyloid hypothesis debate, tau, neuroinflammation

9.5.10 Parkinson's: alpha-synuclein, gut-brain axis

9.6 Cancer Biology Frontier

- 9.6.1 Cancer evolution: phylogenetic reconstruction of tumours
- 9.6.2 Clonal haematopoiesis: precursor to blood cancers
- 9.6.3 Ferroptosis: iron-dependent cell death in cancer
- 9.6.4 Metabolic reprogramming: Warburg effect, glutamine addiction
- 9.6.5 Tumour-microbiome interactions
- 9.6.6 CAR-T for solid tumours: challenges
- 9.6.7 Personalised cancer vaccines: neoantigen prediction
- 9.6.8 Antibody-drug conjugates (ADCs)
- 9.6.9 PROTAC technology: targeted protein degradation

9.7 Immunology and Infection Frontier

- 9.7.1 Innate immune memory: trained immunity
- 9.7.2 Mucosal immunology: gut, lung, skin
- 9.7.3 Microbiome and immunity: FMT, microbial therapeutics
- 9.7.4 Next-generation vaccines: self-amplifying mRNA, mucosal vaccines
- 9.7.5 Pan-coronavirus vaccines: breadth vs depth
- 9.7.6 Broadly neutralising antibodies: HIV, influenza
- 9.7.7 Antimicrobial peptides as new antibiotics
- 9.7.8 Phage therapy: bacteriophages against resistant bacteria

9.8 Open Grand Challenges in Biology and Medicine

- 9.8.1 What is consciousness? Neural correlates and theories
- 9.8.2 Can we cure Alzheimer's disease?
- 9.8.3 What causes ageing and can we reverse it?
- 9.8.4 Why does cancer metastasise and how can we stop it?
- 9.8.5 Can we predict protein-protein interactions from sequence alone?
- 9.8.6 How does the gut microbiome influence brain health?
- 9.8.7 What determines cell fate and how do we control it?
- 9.8.8 Can we grow replacement organs for transplantation?
- 9.8.9 How do we develop universal vaccines for RNA viruses?
- 9.8.10 Can AI accelerate drug discovery to reduce development time and cost?
- 9.8.11 What are the limits of CRISPR in treating genetic disease?
- 9.8.12 How do environmental factors interact with the genome in disease?

COMPLETE ROADMAP -- AT A GLANCE

Chap.	Level	Age	Core Areas	Topics
1	L1	3-7	Living things, plants, animals, body, environment	~40
2	L2	7-11	Cell basics, body systems, photosynthesis, ecosystems, health	~70
3	L3	11-14	Cell biology, genetics, evolution, ecology, human body systems	~95
4	L4	14-16	IGCSE: biochemistry, molecular genetics, microbiology, physiology	~110
5	L5	16-18	IB Level: advanced biochemistry, cell biology, molecular genetics, neuroscience, evolution	~105
6	L6	18-21	BSc/Pre-clinical: metabolic biochemistry, immunology, anatomy, physiology, pharmacology	~95
7	L7	21-24	Clinical medicine, pathology, research methods, neuroscience	~100
8	L8	24-30	PhD/Specialisation: CRISPR, AlphaFold, immunotherapy, systems biology, global health	~90
9	L9	27+	Research frontier: RNA biology, connectomics, cancer evolution, Nobel-level	~85

TOTAL				~850+ Topics
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Yeh roadmap Oxford Biochemistry / Medicine, Cambridge Natural Sciences (Biological), Harvard Medical School, Johns Hopkins, Stanford Medicine, aur Wellcome Sanger / Broad Institute ke research standards par based hai. Plants ki saans se CRISPR tak, phagocytosis se cancer immunotherapy tak -- zindagi ki tamaam bunyadon ko samjho. Medicine mein jaao ya research mein -- dono raaste yahan se shuru hote hain.